

Social Media and Network Analysis
Assignment 2 | Semester 1, 2026

WORKSHEET

Bitcoin Twitter Discussion Analysis

Group Number

Group 57

Team Leader

Batbilegt Baasantsogt (s4042215)

Member 2

Jack Bangit(s4007760)

Member 3

Ashwin Samuel Arul Pragasam (s4055700)

1. Team Project Plan

The following table summarizes the planned work breakdown by days and team member starting from 5th of May. Tasks were assigned based on the three analytical components: network analysis, sentiment analysis, and text/topic analysis.

Day	Batbilegt Network & Data	Jack Bangit Sentiment	Ashwin Samuel Arul Pragasam
May 5	Teams chat created	Joined Teams chat	Joined Teams chat and reviewed the group project requirements.
May 8	Topic selection conversation	Topic selection conversation	Participated in topic selection and supported the decision to analyze our dataset
May 18	Data cleaning and filtering pipeline (4.6M to 632k tweets), Github repository initialization	Dataset exploration	Dataset exploration for text and topic analysis. Reviewed tweet text quality and identified suitable text fields for word frequency and topic modelling
May 19 – 21	independent analysis on network Hashtag co-occurrence network, centrality analysis, community detection	independent analysis on sentiment VADER sentiment analysis, stratified sample RoBERTAa sentiment analysis, produced time series follower-tier, user type plots	independent analysis for topic modelling Cleaned tweet text, performed word frequency and bigram analysis, applied TF-IDF vectorisation, built the NMF topic model, and identified five coherent topics.
May 22-24	Dependent analysis for research question: Mention network, HITS analysis, cross-analysis (network × sentiment × topics)	Dependent analysis for research question: Export of enhanced data frame (sentiment compound + classification), cross-analysis (topic × sentiment)	Dependent analysis for research question: labelled and interpreted the final topics, analysed topic distribution, exported the topic-labelled dataset, and supported topic × sentiment integration with Jack's sentiment analysis.
May 24 – 25	Report integration, final review, submission	Report integration, final review, submission	Report integration, final review, submission

Communication was managed via a shared GitHub repository and group messaging. Weekly check-ins were held to review progress and coordinate integration of the three analytical components.

2. Weekly Timesheets

Batbilegt Baasantsogt (s4042215): Network Analysis & Data Cleaning

Day	Task / Activity	Hours	Cumulative
Day 1	Download dataset, initial exploration, GitHub repo setup	1	1
Day 2	Convert CSV to Parquet, null value removal	2	3
Day 2	Multi-stage filtering pipeline (spam, bots, duplicates)	4	7
Day 2	EDA. account age buckets, tweet volume distribution, temporal plots	2	9
Day 3	Hashtag co-occurrence network construction and threshold selection	5	14
Day 3	Centrality measures (degree, betweenness, PageRank), K-core decomposition	4	18
Day 4	Louvain community detection and community characterization	3	21
Day 4	User mention network construction, in-degree and PageRank analysis	5	26
Day 5	HITS analysis (hubs and authorities)	2	28
Day 6	Cross-analysis: network communities x sentiment x topics	6	36
Day 7-10	Network visualizations, report writing (Sections 2, 3, 6, 7), submission prep	10	46
TOTAL			46

Jack Bangit (s4007760): Sentiment Analysis

Day	Task / Activity	Hours	Cumulative
Day 1	Import dataset parquet, dataset exploration, general setup for sentiment analysis	2	2
Day 2	VADER sentiment analysis, compound score sentiment classification	4	6
Day 3	Sentiment distribution, visualisations	3	9
Day 4	Time-series sentiment analysis, average VADER compound score, visualizations	3	12
Day 5	User-based (verified status) sentiment analysis, follower-tier sentiment analysis, visualizations	3	15
Day 6	RoBERTa sentiment analysis, stratified sampling	7	22
Day 7	Imported enhanced dataset with topics, apply VADER compound scores to dataset	1	23
Day 7	Topic-based sentiment distribution, average VADER compound score per topic, visualisations	4	27
Day 8	Kruskal-Wallis statistical test, export enhanced dataset with topics + sentiment	2	29
Day 8 - 10	Report writing, submission prep	7	36
TOTAL			36

Person 3 Topic: Ashwin Samuel Arul Pragasam (s4055700)

Day	Task / Activity	Hours	Cumulative
Day 1	Reviewed project research question, dataset structure, and topic modelling requirements. Checked text fields and planned the NLP workflow for the topic analysis section.	2	2
Day 2	Conducted text preprocessing for topic modelling, including cleaning tweet text, removing noise, preparing cleaned text for vectorisation, and checking output quality.	4	6
Day 3	Performed word frequency and bigram analysis to identify common Bitcoin discussion terms and early theme patterns before formal topic modelling.	4	10
Day 4	Applied TF-IDF vectorisation to convert cleaned tweet text into numerical features suitable for topic modelling. Tested model settings and reviewed keyword quality.	4	14
Day 5	Built the NMF topic modelling model and selected five final topics. Extracted top keywords for each topic and manually labelled the topics based on keyword meaning.	5	19
Day 6	Analysed topic distribution across the dataset and interpreted which Bitcoin discussion themes appeared most strongly in the Twitter conversation.	4	23
Day 7	Linked topic modelling results with VADER sentiment scores by calculating average sentiment score by topic and identifying which topics had stronger emotional reactions.	3	26
Day 8	Wrote the Topic Modelling report section, prepared tables and figures, added explanations for TF-IDF and NMF, and contributed to the conclusion.	4	30
Day 9	Reviewed topic modelling outputs against the research question, checked consistency of topic labels, and supported final report formatting and integration.	4	34
Day 10	Reviewed final report consistency, checked topic labels, improved explanations, and supported final integration.	2	36
TOTAL			36

3. Work Contribution Summary

The following table summarizes the primary responsibilities and estimated percentage contribution of each team member across the project.

Team Member	Primary Responsibilities	Est. Hours	% Contribution
Batbilegt Baasantsogt (s4042215)	Data cleaning/filtering, EDA, network analysis (hashtag + mention networks, centrality, HITS, community detection, cross-analysis), report Sections 2, 3, 6	46	33.3
Jack Bangit (s4007760)	Text preprocessing for NLP, VADER and RoBERTa sentiment analysis, temporal analysis, sentiment by user type, topic-sentiment linkage, report Section 4	36	33.3
Ashwin Samuel Arul Pragasam (s4055700)	Word frequency and bigram analysis, TF-IDF + NMF topic modelling, topic labelling and validation, report Section 5	36	33.3

4. Individual Self-Reflections

Batbilegt Baasantsogt (s4042215): Network Analysis & Data Cleaning

My primary contribution to this project was the data pipeline and network analysis. I was responsible for building the full filtering pipeline that reduced 4.6 million raw tweets to a quality dataset of 632,179, which the entire team's analysis depended on. I then built both the hashtag co-occurrence and user mention networks, computed centrality measures including degree, betweenness, PageRank, and HITS, and ran Louvain community detection.

What worked well was the modular structure of our notebooks. Each person owning one area meant we could work in parallel without conflicts. The GitHub repository was essential for keeping everything coordinated. I was also pleased with the depth of the cross-analysis section, which connected network communities to sentiment and topics and produced genuinely interesting findings.

What was more challenging than expected was the data quality problem. I underestimated how much noise was in the dataset and the filtering pipeline required significantly more iteration than planned. If I were to do this again, I would start the filtering process earlier and spend more time for exploratory data quality checking before committing to the analytical pipeline. Also, we have chosen a research question that uses all our analysis. In that part we cannot work in parallel, since sentiment analysis on the topics needs a data frame that includes a feature “topic” on all the tweets. And in the network community it needs both topic and sentiment features. This part was very time consuming, which if I were to do it again, I would have planned the research analysis much earlier, since I was the last person in that pipeline.

Jack Bangit(s4007760): Sentiment

My primary contribution was conducting the sentiment analysis for the Bitcoin Twitter dataset. I applied VADER to the full dataset to classify tweets as positive, neutral, or negative, and to generate compound sentiment scores. I also analyzed sentiment across time, verified status, follower tiers, and topics, and used RoBERTa on a stratified sample as a comparison model.

What worked well was the use of different visualizations to show sentiment patterns clearly. The time-series plots helped show how Bitcoin sentiment changed month by month, while the follower tier and topic level charts helped compare sentiment across different groups. The violin plots were especially useful as they showed the spread of compound scores, not just whether tweets were positive, neutral, or negative.

The main challenge was using RoBERTa because it was much and more computationally expensive than VADER. Running it across the full dataset was not practical, so I had to use a stratified sample instead. This required careful sampling across months to avoid skewing the results toward only high-volume periods and to make the RoBERTa comparison more representative.

Person 3 Topic: Ashwin Samuel Arul Pragasam (4055700)

My main contribution was the topic modelling and text analysis section. I cleaned and prepared the tweet text, reviewed common word and bigram patterns, applied TF-IDF vectorisation, and used NMF to identify five clear Bitcoin discussion topics. I also labelled and interpreted the topics, including buying/selling behaviour, Bitcoin as a store of value, crypto projects, Ethereum/altcoins, and market cycles.

What worked well was that the NMF model produced clear and meaningful topics that connected strongly with the project's research question. I also linked the topic results with VADER sentiment scores, which helped show which topics created stronger positive or mixed emotional reactions. This made the topic modelling section more analytical rather than just descriptive.

The main challenge was connecting the technical topic outputs back to the wider network and sentiment findings. To improve this, I added topic distribution and average sentiment score by topic. Next time, I would plan the integration between topics, sentiment, and network analysis earlier so the final cross-analysis is smoother and stronger.